

Alex S. Huang

CONTACT INFORMATION	<i>Phone:</i> (480) 749-8123 <i>E-mail:</i> alex.huang@utdallas.edu <i>Website:</i> https://alexhuang1029.github.io/	
RESEARCH INTERESTS	VLAs/WAMs, robot learning, RL/ML, benchmarking, assistive robotics, prosthetics, manipulation.	
EDUCATION	The University of Texas at Dallas , Richardson, Texas, USA B.S., Electrical Engineering, 2025 - 2028 (<i>Expected</i>) GPA: 4.00/4.00 Hobson Wildenthal Honors College	
EXPERIENCE	The University of Texas at Dallas , Richardson, Texas USA <i>Research Assistant, Intelligent Robotics and Vision Lab (IRVL)</i> 08/2025 - present Researching robot learning, VLAs, benchmarking, and assistive robotics. <i>Treasurer, Accessible Prosthetics Initiative</i> 05/2026 - present Manage funding & sponsorships for 501(c) designing low-cost prosthetics for amputees. <i>Associate Philanthropy Chair, Pi Kappa Phi Fraternity</i> 09/2026 - present Organize service events & fundraising for Pi Kappa Phi's philanthropy, The Ability Experience. <i>Eta Kappa Nu Member, IEEE UTD</i> 09/2025 - present Nominated for Eta Kappa Nu Honor Society for the Institute of Electrical and Electronics Engineers. Oregon State University , Corvallis, Oregon USA <i>Undergraduate Research Intern, Do Robotics Lab</i> 06/2026 - 08/2026 1 of 14 undergraduates selected nationwide for Robots in the Real World REU program. Arizona State University , Tempe, Arizona USA <i>Research Assistant, Sun Robotics Lab</i> 09/2024 - 08/2025 High school research intern, designed novel, cost-effective robotics software. <i>Software Engineering Intern, SEFCOM Lab</i> 05/2024 - 08/2024 High school summer intern, developed cybersecurity software to counteract SMS phishing. Stanford University , Palo Alto, California USA <i>Research Assistant</i> 06/2024 - 08/2024 Selective (5%) internship with acclaimed CS & AI faculty, investigated LLM prompt engineering.	
HONORS AND AWARDS	NSF Computing Research Association CISE Undergraduate Research Award (\$10,000) 09/2026 NSF Award #2520553 Phi Kappa Phi Honor Society 09/2026 2025-26 Dean's List 09/2026 Robots in the Real World REU Student Award funding (\$9,000) 06/2026 NSF Award #2349747 UT Dallas Undergraduate Research Scholar Award (\$800) 04/2026 UT Dallas Full-ride National Merit Scholarship (valued at \$200k+) 08/2025 Regeneron 2025 International Science & Engineering Fair Finalist 05/2025 Cybersecurity & Trusted Foundations High School Internship Grant (\$2,500) 07/2024	

- PUBLICATIONS
- [1] **A Multimodal Amphibious Tracked Robot**
Liv Dawes, **Alex S. Huang**, Brian H. Do
In submission, *Science Robotics*, 2026.
 - [2] **VLA-REPLICA: A Low-Cost, Reproducible Benchmark for Real-World Evaluation of Vision-Language-Action Models**
Alex S. Huang*, Jiahui Zhang*, Shiqing Tang, Yu Xiang
In *Advances in Neural Information Processing Systems (NeurIPS)*, 2026.
In *Data-Centric Robotics Workshop, Robotics: Science and Systems (RSS)*, 2026.
 - [3] **Robometer: Scaling General-Purpose Robotic Reward Models via Trajectory Comparisons**
Anthony Liang*, Yigit Korkmaz*, Jiahui Zhang, Minyoung Hwang, Abrar Anwar, Sidhant Kaushik, Aditya Shah, **Alex S. Huang**, Luke Zettlemoyer, Dieter Fox, Yu Xiang, Anqi Li, Andreea Bobu, Abhishek Gupta, Stephen Tu, Erdem Biyik, Jesse Zhang
In *Robotics: Science and Systems (RSS)*, 2026.
 - [4] **THETA: Triangulated Hand-State Estimation for Teleoperation and Automation in Robotic Hand Control**
Akshay Karthik*, **Alex S. Huang***
In *2025 International Conference on Engineering and Emerging Technologies (ICEET)*, 2025.
- MENTORSHIP
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|---|--|-------------------|
| Hanna Nguyen, Undergraduate Student (EE), UT Dallas | | 04/2026 - present |
| May Lynn Espinola, Undergraduate Student (CS), UT Dallas | | 08/2026 - present |
| Charita Sree Potluri, Undergraduate Student (EE), UT Dallas | | 09/2026 - present |
- TALKS & PRESENTATIONS
- [1] **VLA-REPLICA: A Low-Cost, Reproducible Benchmark for Real-World Evaluation of Vision-Language-Action Models**
Advances in Neural Information Processing Systems (NeurIPS), 12/2026 (*Upcoming*)
 - [2] **A Multi-modal Paddlewheel Mechanism for Amphibious Robot Propulsion**
Robots in the Real World REU Symposium
Oregon State University, 08/2026.
 - [3] **VLA-REPLICA: A Low-Cost, Reproducible Benchmark for Real-World Evaluation of Vision-Language-Action Models**
The Data-Centric Robotics Workshop
Robotics: Science and Systems (RSS), 07/2026.
 - [4] **VLA-Replica: A Low-Cost, Replicable Benchmark to Evaluate Robotic Vision-Language-Action Models**
UT Dallas Undergraduate Research Poster Symposium
The University of Texas at Dallas, 04/2026.
 - [5] **THETA: Triangulated Hand-State Estimation for Teleoperation and Automation in Robotic Hand Control**
2025 International Conference on Engineering and Emerging Technologies, 10/2025.
 - [6] **THETA: Triangulated Hand-State Estimation for Teleoperation and Automation in Robotic Hand Control**
2025 Regeneron International Science & Engineering Fair, 05/2025.
 - [7] **Optimizing Large Language Models: Learning from Mistakes in Gameplay**

2024 STEM2SHTeM Research Symposium
Stanford University, 08/2024.

- [8] **SmishSmashing: A Software Utilizing LLMs to Detect and Respond to Phishing Scams**
The Cybersecurity & Trusted Foundations Summer Internship Symposium
Arizona State University, 07/2024.
- [9] **NeuroTrace: A Novel System to Detect Neurodegeneration through Handwriting Kinematics Analysis**
Arizona Science & Engineering Fair, 04/2024.